



www.ijcci.info

International Journal of Clinical Cases and Investigations

Case Study

Percutaneous transluminal angioplasty and stenting of celiac artery stenosis with thrombus in the treatment of mesenteric angina

S S Prakash, Jimmy George, Sree Ranga P C, Shivakumar Byrappa, S Shankar, C N Manjunath

Sri Jayadeva Institute of Cardiovascular Sciences & Research, Bangalore, India

Email: sreerangapallaki@gmail.com

International Journal of Clinical Cases and Investigations. Volume 6 (Issue 3), 43:45, 1st April 2015.

Abstract

Mesenteric angina, also known as chronic mesenteric ischemia (CMI) or intestinal angina is a condition characterised by inadequate blood supply to bowel resulting from stenosis of one or more of the three mesenteric arteries. The standard therapeutic option is surgical (bypass or endarterectomy). Endovascular treatment was first done in 1980 & can be used for focal lesions or proximal involvement of vessel. We present a case of a 37 years old man presenting with occlusion of IMA, 90% stenosis of SMA and 75% stenosis of celiac artery with thrombus treated with percutaneous transluminal angioplasty and stenting.

Keywords: Percutaneous transluminal angioplasty, celiac artery stenosis, mesenteric angina

Introduction

Mesenteric angina, also known as chronic mesenteric ischemia (CMI) or intestinal angina is an uncommon and difficult diagnosis to make. CMI is a condition characterised by inadequate blood supply to bowel resulting from stenosis of one or more of the three mesenteric arteries; celiac artery, superior mesenteric artery & inferior mesenteric artery. Atherosclerosis is the most common cause of stenosis & females are affected three times more than males. Symptomatic untreated bowel ischemia may cause malnutrition, acute bowel ischemia with infarction that is associated with bad prognosis. The standard therapeutic option is surgical (bypass or endarterectomy). Endovascular treatment was first done in 1980 & can be used for focal lesions or proximal involvement of vessel.

Case Report

A 37 years old gentleman, presented with postprandial colicky abdominal pain for past 6 months. This was associated with weight loss of about 20Kg in past 6 months. His past medical history revealed he is a smoker with ischemic heart disease (old AAMI). Extensive gastro intestinal investigations were done in last 6 months & were normal. A C T angiogram of abdomen was done showing celiac and Superior mesenteric artery stenosis. In view of patients past medical history and C T finding, a provisional diagnosis of mesenteric angina was made and planned to proceed for mesenteric angiography and stenting. This was carried out and revealed occlusion of IMA, 90% stenosis of SMA and 75% stenosis of celiac artery with thrombus. In view of convincing history of CMI, weight loss, it was decided to proceed to PTA and stenting.

Method

The procedure was approached accessing via right femoral A and left brachial A approach. A 6 F short sheath was inserted to right femoral artery and another 6F hydrophilic sheath inserted through left brachial artery after skin infiltration of lignocaine LA, Heparin 5000 IU IV was given . The femoral sheath was used for monitoring and non-selective angiography, while brachial sheath for delivering balloon and stent. A 4mm PTCA balloon was passed over 0.014 guide wire and dilatation of ostial & proximal segment of celiac artery done.

A 5.0 X 12mm DES stent mounted on a balloon was inserted and inflated just below burst pressure and kept for 30 seconds. An angiography conducted after the procedure showed a diameter equal to the nearby normal segment. Guide wire could not be passed through Superior mesenteric artery, hence could not be stented and was left. Brachial sheath was removed immediately and haemostasis secured, femoral sheath removed after 4 hours. There was no post-operative complications. Patient was discharged on the 3rd day with aspirin and clopidogrel , subsequent follow up after 2 weeks /2 months and 6 months no postprandial abdominal pain .Patient gained 10 kg weight in 6 months.

Discussion

Mesenteric angina occurs when stenosis or occlusion of the mesenteric artery limits blood flow to the intestine. Majority of cases are due to atherosclerosis, other causes include antiphospholipid antibody syndrome, Fabry's disease, Behcet's disease, thromboangitis obliterans, Takayasu arteritis, Crohn's disease and external compression. The failure to provide an increase in gastrointestinal blood flow, following food intake due to underlying narrowing of mesenteric artery leads to gastrointestinal ischemia and the onset of symptoms of intestinal angina.

The aims of therapy are to: 1) achieve symptomatic resolution, 2) prevent bowel infarction and resultant morbidity and mortality and 3) prevent and treat malnutrition. Some groups also recommend prophylactic revascularization even in asymptomatic patients who are due for an operation in the aorta (e.g. aneurysm, coarctation, etc.) or before being scheduled for major abdominal surgery that might jeopardize the collateral circulation

References:

1. Sharafuddin MJ, Olson CH, Sun S, *et al.*: Endovascular treatment of celiac and mesenteric arteries stenoses: application and results. *J Vasc Surg.* 2003; 38(4): 692–8
2. Furrer J, Grüntzig A, Kugelmeier J, *et al.*: Treatment of abdominal angina with percutaneous dilatation of an arteria mesenterica superior stenosis. Preliminary communication. *Cardiovasc Intervent Radiol.* 1980; 3(1): 43-4
3. Rose SC, Quigley TM, Raker EJ: Revascularisation for chronic mesenteric ischemia: comparison of operative arterial bypass grafting and percutaneous transluminal angioplasty. *J Vasc Interv Radiol.* 1995; 6(3): 339–49
4. Allen RC, Martin GH, Rees CR, *et al.*: Mesenteric angioplasty in the treatment of chronic intestinal ischemia. *J Vasc Surg.* 199

Figures